

Interpreting HARP v4.1 graphical output

Introduction

HARP v4.1 (Heptad Assignment Register Probe) was developed to examine the molecular interface between the myosin-tail-like coiled-coil region (MyhT) and **IBAM (C12orf29; In Between Actin and Myosin)**, specifically where MyhT binds within the **major groove (MG) of IBAM**. HARP asks whether the MyhT residues forming this interface are organized with respect to an independently defined MyhT heptad register.

The important distinction is that HARP does not derive the heptad register from the molecular-dynamics contact pattern itself.

Instead, two independent sources of information are brought together:

DeepCoil2 (DC2) (Gabler et al., 2020; Zimmermann et al., 2018) provides the sequence-based coiled-coil register and therefore the assignment of MyhT residues to the canonical *a-g* heptad coordinate system.

The **GROMACS molecular-dynamics trajectory** (Abraham et al., 2015) (`md.xtc`, interpreted with its corresponding topology) independently provides the three-dimensional behaviour of the MyhT–IBAM complex through time, from which HARP determines which MyhT residues contact IBAM and how persistently those contacts occur.

HARP then asks whether these trajectory-derived MyhT–IBAM contacts show structure relative to the independently defined DC2 heptad coordinate system.

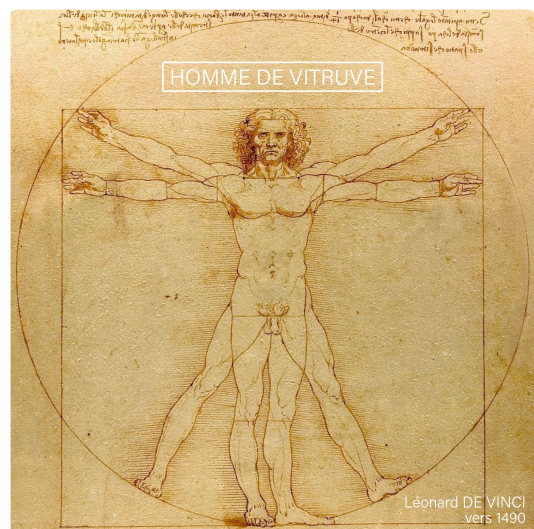
The six principal graphical outputs therefore form a progression:

trajectory → contact profile → heptad phase occupancy → temporal persistence → statistical testing

No individual figure should be interpreted in isolation as proof of a heptad-organized interface. The strength of HARP comes from the agreement—or disagreement—among these complementary views of the same trajectory-derived contact data. Here, we compare HARP outputs from *Ischnura elegans* and *Homo sapiens* MyhT–IBAM complexes.



Ischnura elegans nymph (Wikimedia)



Leonardo da Vinci, *Vitruvian Man*.

Figure 01 — Trajectory-derived MyhT contact profile

What question does this figure answer?

Where along the MyhT sequence does the MG actually make contact during the MD trajectory?

This is the most direct representation of the trajectory-derived interaction and is therefore the natural starting point for interpreting a HARP analysis.

Before examining the trajectory-derived contact profiles, it is useful to see the independent DeepCoil2 predictions from which the MyhT heptad registers are obtained. DeepCoil2 defines the coiled-coil interval and assigns the repeating *a* and *d* core positions that anchor the HARP heptad coordinate system. Importantly, this assignment is made independently of the molecular-dynamics contact data subsequently analysed by HARP.

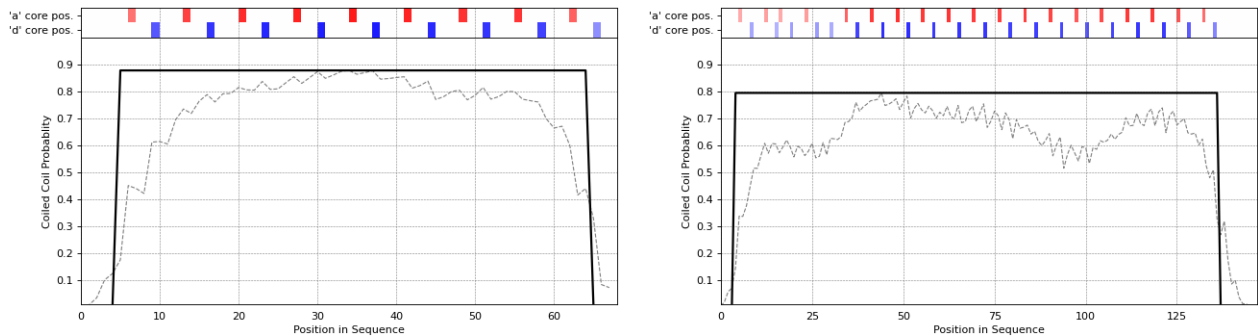


Figure 1: DeepCoil2 coiled-coil predictions for *Ischnura* (left) and *Homo* (right). The CC-Prob profiles define the predicted coiled-coil regions, while the *a* and *d* core-position assignments provide the independently predicted heptad register used by HARP. The two taxa differ substantially in coiled-coil length and probability profile, but in both cases the register is established before trajectory-derived MyhT-IBAM contact information is considered. DeepCoil2 therefore supplies the coordinate system onto which HARP subsequently projects the observed molecular-dynamics contacts.

HARP then asks a different question: given this independently assigned heptad register, where along MyhT does the MyhT-IBAM interface actually occur during the molecular-dynamics trajectory?

Axes

The x-axis represents MyhT sequence position.

The y-axis, labelled MG contact occupancy, represents the fraction of the analysed trajectory during which a MyhT residue satisfies the HARP contact criterion.

Thus:

- 0.0 means that the residue essentially never participates in an MG contact;
- 0.5 means that contact is present for approximately half of the analysed observations;
- 1.0 means that contact is present essentially throughout the analysed trajectory.

The shaded region denotes the DC2-defined coiled-coil region used for register analysis. The *a* and *d* annotations beneath the sequence provide visual landmarks for the independently assigned heptad coordinate system.

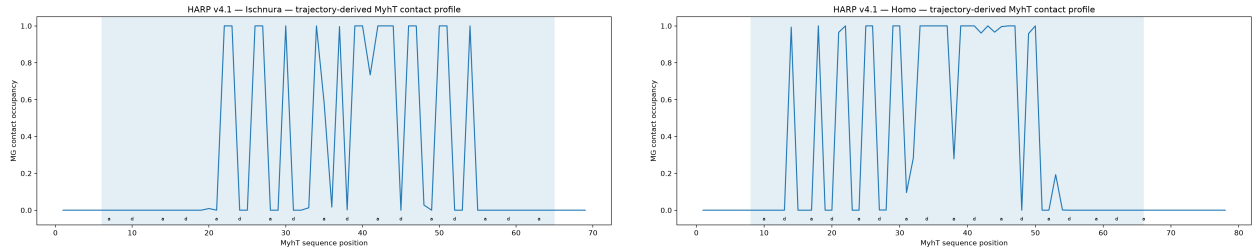


Figure 2: Trajectory-derived MyhT contact profiles for *Ischnura* and *Homo*. MyhT-IBAM major-groove (MG) contact occupancy is plotted as a function of MyhT sequence position for representative *Ischnura* (left) and *Homo* (right) trajectories. The shaded region denotes the DeepCoil2-defined coiled-coil interval, with the predicted *a* and *d* heptad positions shown along the sequence axis. Contact occupancy ranges from 0 (no MG contact during the sampled trajectory) to 1 (persistent contact), thereby converting the molecular-dynamics trajectory into a residue-resolved description of where MyhT engages the IBAM major groove. Both taxa show highly non-uniform contact profiles, with contacts concentrated within particular portions of the DeepCoil2-defined interval rather than distributed uniformly along MyhT. Importantly, the detailed profiles differ between *Ischnura* and *Homo*, illustrating that HARP begins with taxon-specific trajectory-derived contact behaviour rather than imposing a common interaction pattern. These residue-level profiles provide the empirical input that is subsequently projected onto the DeepCoil2-anchored *a-g* heptad coordinate system for phase-level analysis.

How to read the figure

A flat profile near zero would indicate little or no persistent interaction.

A broad elevated region would indicate an extended contact surface.

Sharp peaks separated by low-occupancy positions indicate a more spatially selective interface in which particular MyhT residues participate much more strongly than their neighbours.

For example, the *Ischnura* profile shown during development contains a strongly structured region in which several residues approach occupancy 1 while adjacent positions can approach zero. *Homo* instead contains broader stretches of high occupancy.

These are therefore clearly different contact architectures even before any heptad statistics are calculated.

What this figure does not establish

A visually repetitive contact pattern must not automatically be called “heptad periodicity.”

Human pattern recognition is extremely good at seeing apparent periodic structure. Figure 2 shows **where contacts occurred**; the subsequent HARP analyses determine whether those contacts have meaningful organization relative to the independently assigned heptad register.

In other words:

Figure 01 shows the phenomenon. It does not by itself establish its statistical explanation.

Figure 02 — Contact occupancy by heptad phase

What question does this figure answer?

When the trajectory-derived contacts are projected onto the DC2 register, which heptad phases carry the interaction?

Every residue in the analysed coiled-coil region belongs to one of seven heptad phases:

a, b, c, d, e, f, g

Figure 02 collapses the sequence-position information from Figure 01 into this seven-phase coordinate system.

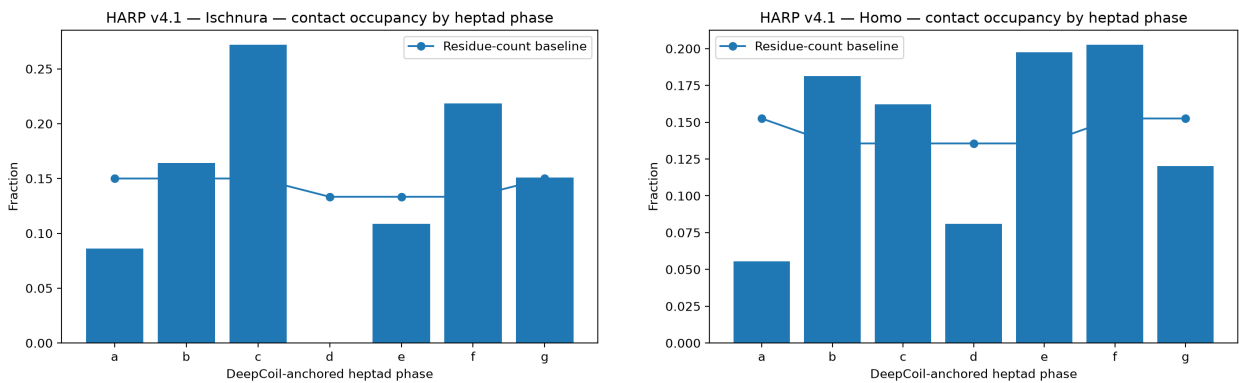


Figure 3: DeepCoil2-anchored contact occupancy by heptad phase for *Ischnura* and *Homo*. Residue-level MyhT-IBAM major-groove contacts from the molecular-dynamics trajectory are collapsed onto the seven DeepCoil2-defined heptad phases (a–g) for *Ischnura* (left) and *Homo* (right). Bars show the fraction of observed MG contact assigned to each phase, while the line indicates the residue-count baseline expected from the number of MyhT residues belonging to each phase. Consequently, enrichment above the baseline indicates that a phase contributes more contact than expected simply from its representation in the analysed sequence, whereas depletion indicates less contact than expected. Both taxa exhibit strongly non-uniform phase occupancy, but with distinct distributions: *Ischnura* shows pronounced enrichment at c and f, with essentially no occupancy at d, whereas *Homo* shows elevated occupancy across b, c, e, and f, with d strongly depleted. Thus, Figure 02 transforms the residue-resolved observations of Figure 01 into a common a–g coordinate system, permitting contact organisation to be compared between MyhT sequences of different lengths and residue composition. This phase representation forms the basis for HARP’s subsequent tests of heptad-phase structure and enrichment.

Bars: observed contact occupancy

The bars show the fraction of trajectory-derived contact occupancy associated with each heptad phase.

A perfectly phase-neutral interaction would not necessarily produce identical bars, however, because the analysed sequence may not contain exactly equal numbers of residues belonging to every phase.

That is why HARP also displays the baseline.

Line: residue-count baseline

The line labelled Residue-count baseline B_k represents the fraction expected from the number of available residues assigned to each phase.

This distinction is essential. Suppose 15% of the available residues belong to phase c, but 27% of the observed contact occupancy occurs at c. The interesting quantity is not merely that c has many contacts; it is that c receives **more contact than would be expected from its representation in the analysed sequence**.

Conversely, a phase falling substantially below its baseline is relatively depleted.

How to read the figure

Look primarily at the difference between each bar and its corresponding baseline, not merely at absolute bar height.

For the example *Ischnura* result, phase c is strongly enriched relative to its residue-count baseline, f is also elevated, whereas a, d, and e are depleted to varying degrees.

Homo displays a different distribution, with strong occupancy across b, e, and f.

This immediately demonstrates an important feature of HARP: different taxa need not achieve their MyhT–MG interaction through identical residue-level contact architectures.

What this figure does not establish

The phase summary collapses information over the trajectory. It therefore cannot tell us whether the observed phase preference was persistent or whether it arose from a short-lived episode.

That question belongs to Figure 03.

Figure 03 — Temporal persistence of phase occupancy

What question does this figure answer?

Does the heptad-phase contact pattern persist through the trajectory, or is the overall pattern being produced by a transient event?

This heatmap adds time back into the analysis.

Axes and colour scale

The x-axis contains the seven DC2-defined heptad phases, a–g.

The y-axis contains successive trajectory blocks.

Colour represents the fraction of contacts assigned to a particular phase during each block. Brighter colours indicate greater contact occupancy; darker colours indicate less.

Each horizontal row can therefore be thought of as a miniature phase profile for one portion of the trajectory.

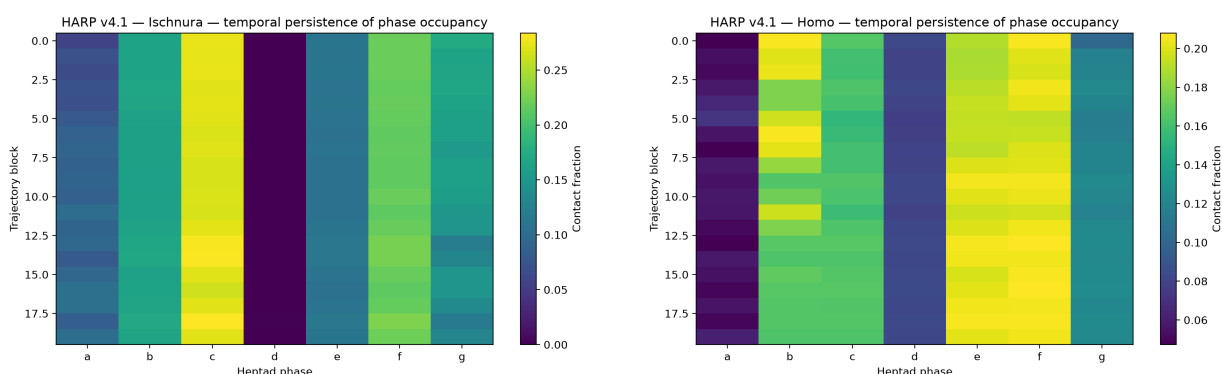


Figure 4: Temporal persistence of DeepCoil2-anchored heptad-phase contact occupancy in *Ischnura* and *Homo*. The molecular-dynamics trajectory is divided into sequential time blocks (rows), and MyhT-IBAM major-groove contact occupancy is recalculated for each DeepCoil2-defined heptad phase (a–g, columns). Colour represents the fraction of contact assigned to each phase within that trajectory block, allowing the phase preferences summarised in Figure 02 to be examined through time rather than only as trajectory-wide averages. *Ischnura* (left) exhibits a remarkably persistent pattern dominated by phase c, with substantial f occupancy and essentially absent d occupancy throughout the trajectory. *Homo* (right) displays a broader distribution, principally involving b, c, e, and f, while a and d remain comparatively depleted. Although local fluctuations occur between trajectory blocks, the major phase preferences remain recognizable throughout the simulation. The heatmaps therefore demonstrate that the non-uniform phase distributions in Figure 02 are temporally persistent features of the trajectory rather than averages produced by a small number of transient contact events. Importantly, the different colour scales should be read independently: colour intensity is used to reveal temporal structure within each taxon and should not by itself be used for direct quantitative comparison between taxa.

How to read the heatmap

Three broad behaviours are possible.

If bright and dark regions jump dramatically between columns as one moves down the heatmap, the interface is temporally unstable with respect to heptad phase.

If particular columns remain consistently bright and others consistently dark, the phase preference is persistent through time.

If the heatmap is relatively uniform across all seven columns, the interaction shows little phase selectivity.

The *Ischnura* example is particularly clear. Phase c remains strongly occupied across essentially the entire trajectory, f remains strongly represented, while d remains effectively absent.

That is important because the phase-summary result is not being generated by one brief interval. The same broad phase architecture repeatedly appears in successive trajectory blocks.

Homo again shows a different but similarly persistent pattern, with strong occupancy distributed principally across b, e, and f.

Relationship to Figure 02

Figures 02 and 03 deliberately examine closely related information from different angles.

Figure 02 asks:

What is the overall phase distribution?

Figure 03 asks:

Was that distribution maintained through time?

Consequently, a strong visual correspondence between them is expected when the phase architecture is temporally stable.

Their agreement is informative rather than redundant.

Figure 04 — Null distribution: period-7 maximum mode

What question does this figure answer?

Does the trajectory-derived contact pattern contain unusually strong seven-residue periodic organisation?

Here HARP moves from description to statistical testing.

The histogram

The histogram represents the statistic obtained from repeated randomized/null realizations.

This is the distribution of results HARP can obtain when the relevant relationship is disrupted according to the specified null procedure.

The dashed vertical line is the statistic calculated from the **actual trajectory-derived data**.

The fundamental question becomes extremely simple:

Where does reality fall relative to the distribution generated under the null?

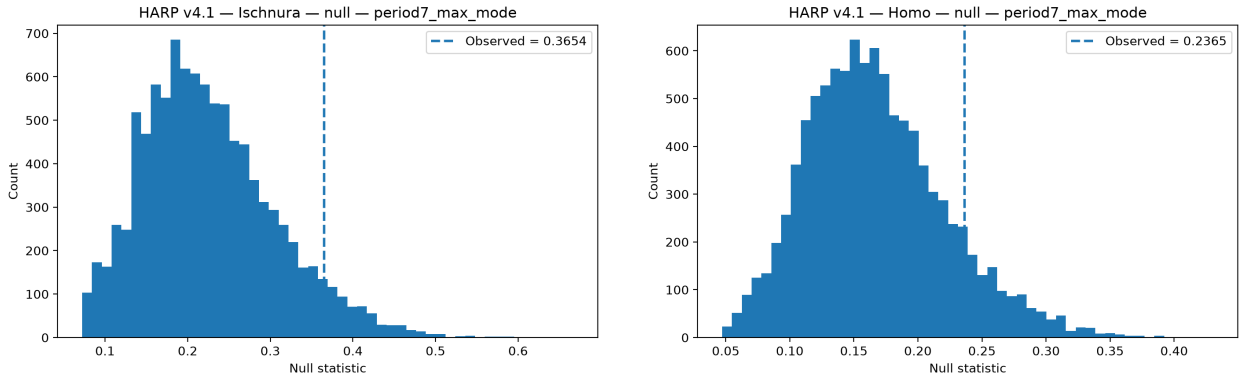


Figure 5: Permutation test for seven-residue periodic organisation of MyhT–IBAM contact occupancy in *Ischnura* and *Homo*. Histograms show the null distributions obtained after repeatedly disrupting the correspondence between MyhT sequence position and the DeepCoil2-anchored heptad register while preserving the underlying contact information. The dashed vertical line marks the `period7_max_mode` statistic measured from the unpermuted trajectory-derived data. This statistic asks whether the distribution of MyhT–IBAM contacts contains a stronger seven-residue periodic component than would normally arise after the register relationship has been randomized. In *Ischnura* (left), the observed statistic (0.3654) lies well into the upper tail of the null distribution, indicating pronounced seven-residue periodic organisation. *Homo* (right) likewise shows an observed statistic (0.2365) displaced toward the upper tail, although the separation from its null distribution is less extreme. Thus, both taxa contain evidence that contact occupancy is structured according to the expected heptad periodicity rather than being distributed arbitrarily along the MyhT sequence. The figure tests periodicity, not whether a particular heptad phase is preferentially occupied; the latter is addressed separately by the phase-specific analyses.

How to read it

If the dashed observed line lies well inside the dense part of the null distribution, the observed value is not particularly unusual under that null model.

If it lies deep in the relevant tail, relatively few randomized realizations reproduce an equally extreme result.

The associated empirical permutation probability provides the formal numerical assessment; the plot provides the visual intuition.

Why “maximum mode”?

A seven-residue periodic signal can occur at different phase offsets. HARP therefore does not require the signal to begin at an arbitrarily chosen sequence coordinate.

The statistic captures the strongest relevant period-7 organization under the implemented analysis.

The dashed line again represents the observed trajectory-derived value; the histogram represents the corresponding null distribution.

A crucial caution

The direction of “interesting” depends upon the definition of the statistic. One should therefore **not adopt the universal rule that farther right always means stronger evidence**.

The appropriate tail and interpretation must follow the statistic implemented by HARP.

This is particularly important when visually comparing different HARP null plots.

Figure 05 — Null distribution: phase non-uniformity

What question does this figure answer?

How strongly unequal is the observed distribution of contacts across the seven heptad phases, and is that degree of non-uniformity unusual under the null model?

This test addresses a broader question than target-phase enrichment.

Rather than asking specifically whether one predefined phase relationship is enriched, it asks whether the seven-phase contact distribution is **more structured/non-uniform than expected**.

Biological intuition

Imagine two hypothetical interfaces.

The first distributes contacts approximately equally:

$$a \approx b \approx c \approx d \approx e \approx f \approx g$$

The second strongly favours a subset:

$$a \text{ low} - b \text{ moderate} - c \text{ high} - d \text{ absent} - e \text{ low} - f \text{ high} - g \text{ moderate}$$

The second has much greater phase non-uniformity.

Figure 05 asks whether the non-uniformity actually observed in the trajectory exceeds what the null model commonly produces.

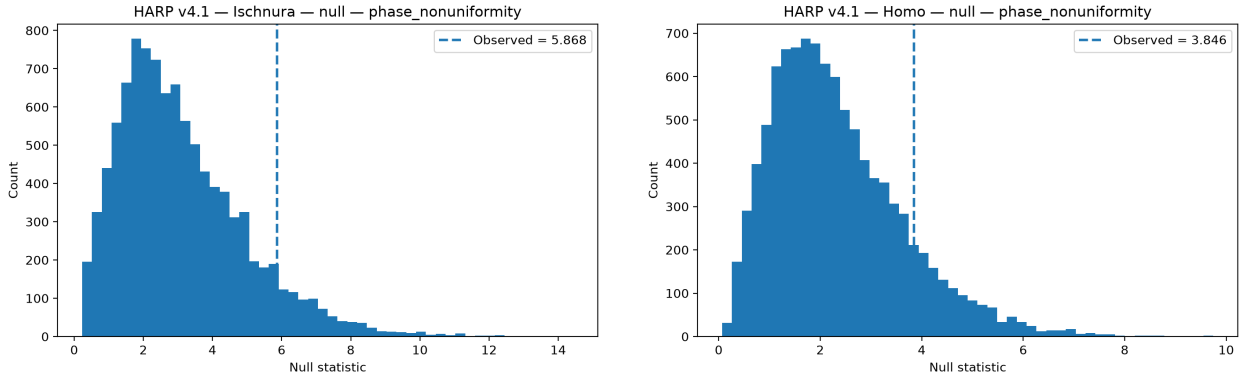


Figure 6: Permutation test for non-uniform distribution of MyhT–IBAM contacts among DeepCoil2-anchored heptad phases in *Ischnura* and *Homo*. Histograms show the null distributions obtained after repeatedly disrupting the relationship between the trajectory-derived MyhT contact pattern and its DeepCoil2-defined heptad register. The dashed vertical line indicates the `phase_nonuniformity` statistic calculated from the observed, unpermuted data. This statistic measures how strongly contact occupancy is concentrated in particular heptad phases rather than being distributed approximately evenly among phases a–g. *Ischnura* (left) has an observed non-uniformity statistic of 5.868, lying well toward the upper tail of its null distribution. This reflects the strongly unequal phase occupancy seen in Figures 02–03, particularly the enrichment of phase c and depletion of phase d. *Homo* (right) also exhibits greater phase non-uniformity than is typical under the null model, with an observed statistic of 3.846, although its displacement is less pronounced than that of *Ischnura*. These results demonstrate that the phase-specific contact distributions are more uneven than would ordinarily be expected after the correspondence between contact position and heptad register has been disrupted. This test evaluates how unevenly contacts are distributed among the seven phases; it does not by itself specify which phase is preferred or test seven-residue periodicity.

Why this is useful

This protects HARP against an overly narrow interpretation of coiled-coil organization.

A biologically structured interface need not necessarily obey the simplistic expectation:

“contacts should occur predominantly at a and d.”

The interaction may instead prefer some other reproducible combination of phases.

Phase non-uniformity therefore asks the more general question:

Does the interface care about heptad phase at all?

before imposing a more specific biological interpretation upon which phases are preferred.

Figure 06 — Null distribution: target-phase enrichment

What question does this figure answer?

Is the observed enrichment at the predefined target heptad phases unusual relative to the randomized null model?

Figure 06 asks a more specific question than the preceding tests. Rather than asking whether the contact pattern shows seven-residue periodic organisation in general, it asks whether contacts are preferentially associated with the predefined target heptad phases.

HARP calculates the `target_phase_enrichment` statistic from the observed trajectory-derived contact pattern expressed in the DeepCoil2-anchored heptad register, then compares that value with the distribution generated under the corresponding randomized null model.

This distinction is important because a contact pattern may be strongly heptad-organised or phase non-uniform without being enriched at the particular target phases specified by the test.

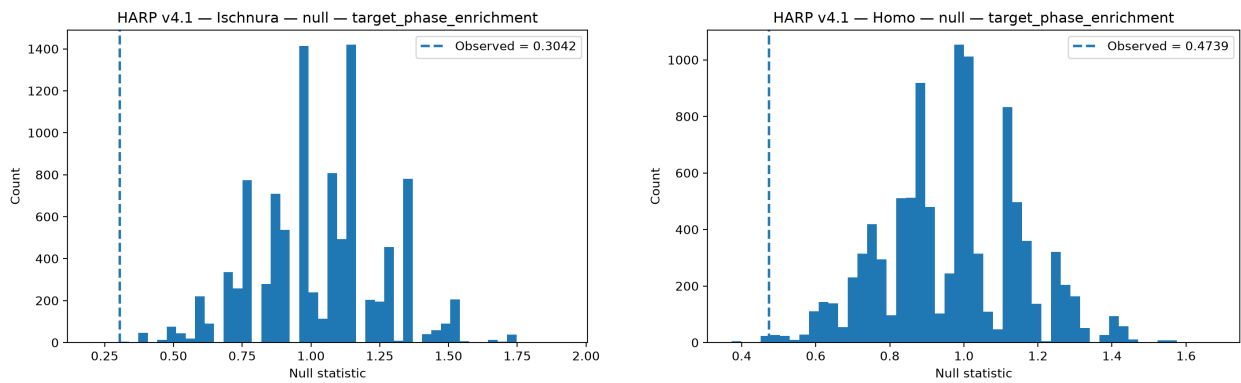


Figure 7: Permutation test for enrichment of MyhT–IBAM contacts at the predefined target heptad phases in *Ischnura* and *Homo*. Histograms show the null distributions generated by repeatedly disrupting the correspondence between the trajectory-derived MyhT contact pattern and its DeepCoil2-anchored heptad register. The dashed vertical line represents the `target_phase_enrichment` statistic calculated from the observed, unpermuted data. Unlike the preceding tests, which ask whether contacts exhibit seven-residue periodicity (Figure 04) or are distributed non-uniformly among heptad phases (Figure 05), this test asks whether contacts are specifically enriched at the predefined target phase relationship. In both *Ischnura* (left; observed = 0.3042) and *Homo* (right; observed = 0.4739), the observed statistic lies toward the lower extreme of the corresponding null distribution. Thus, although both taxa exhibit structured and non-uniform heptad-associated contact patterns, that organisation does not take the form of enrichment at the particular target phases represented by this statistic. Figure 06 therefore distinguishes general heptad organisation from conformity to a specific hypothesised phase arrangement.

How to read it

An observed value embedded within the bulk of the null distribution indicates that comparable target-phase enrichment is readily produced under the null.

An observed value in the relevant extreme of the distribution indicates that the observed target-phase relationship is unusual relative to the randomized controls.

The empirical permutation probability should be used for formal inference rather than estimating significance by eye.

How the six figures fit together

The easiest way to understand HARP is not as six independent statistical tests but as a chain of increasingly abstract questions about the same molecular interaction.

Figure	Question	Level of analysis
01 Contact profile	Where along the MyhT sequence are contacts with IBAM observed?	Residue / trajectory
02 Phase occupancy	How are those contacts distributed among the DeepCoil2-anchored heptad phases?	Register
03 Temporal persistence	Does that phase-associated contact pattern persist through the trajectory?	Dynamics
04 Period-7 null	Does the trajectory-derived contact pattern contain unusually strong seven-residue periodic organisation?	Statistical
05 Non-uniformity null	Is contact occupancy distributed among the seven heptad phases more unevenly than expected under the null model?	Statistical
06 Target-phase null	Are contacts specifically enriched at the predefined target heptad phases relative to the randomized null model?	Statistical

Together, these figures move from direct trajectory-derived observation to progressively more abstract statistical tests of heptad-associated organisation.

What HARP v4.1 is actually testing

HARP brings together two independently derived sources of information: the DeepCoil2-defined MyhT heptad register and the GROMACS trajectory-derived pattern of contacts between MyhT and the major groove (MG) of IBAM (C12orf29). The former provides the sequence-based coiled-coil coordinate system (*a-g*), while the latter describes where and how persistently MyhT contacts the IBAM major groove during the molecular dynamics trajectory.

HARP maps these trajectory-derived MyhT-IBAM MG contacts onto the independently assigned MyhT heptad coordinate system and tests whether the resulting organization is greater than expected under the specified null models.

Importantly, the MD contact pattern is not used to define the heptad register against which it is subsequently tested. The two sources of information therefore remain independent at the point at which HARP brings them together.

The central question addressed by HARP can therefore be stated simply:

When independently trajectory-derived contacts between MyhT and the IBAM major groove are expressed in an independently established MyhT coiled-coil coordinate system, is their distribution compatible with random placement, or does reproducible heptad-associated organization emerge?

Figures 01–06 are six complementary views of that question.

What HARP does not demonstrate

This deserves an explicit section in the repository because it heads off several possible overinterpretations.

HARP does not, by itself, prove that a particular residue makes a particular biochemical interaction in vivo.

It does not prove that an AF-derived complex represents the unique physiological configuration.

It does not establish causality from statistical periodicity.

It does not mean that every taxon must exhibit the same phase architecture.

And a significant period-7 or phase statistic does not magically convert an uncertain structural model into biological fact.

Rather, HARP tests a narrower and more defensible proposition:

Given the analysed molecular model and trajectory, are the observed MyhT–MG contacts organized, with respect to the independently assigned MyhT heptad register, to a degree that is unusual under the specified null model?

Panel null — cross-taxon shared phase alignment

Purpose

The HARP v4.1 panel analysis evaluates whether independently analysed taxa exhibit a **shared organization of MyhT–MG contacts with respect to the DeepCoil2-anchored heptad coordinate system**.

It is important to distinguish the panel analysis from a simple aggregation of the taxon-level graphical outputs.

The panel statistic does not numerically combine the contact-profile plots, temporal heatmaps, and taxon-level null distributions. Those figures represent complementary analyses and visualizations performed at the individual-taxon level.

Instead, the panel analysis operates on the **taxon-level phase representation** derived from the underlying trajectory/contact analysis.

Conceptually:

**MD trajectory → residue contacts → DC2 heptad phases → taxon phase profile →
cross-taxon comparison**

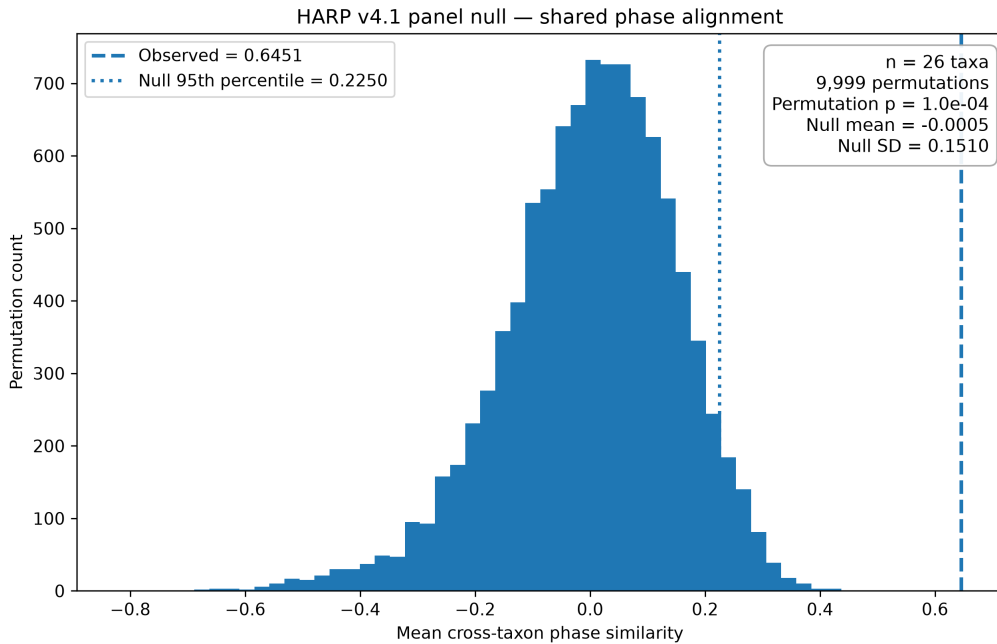


Figure 8: Schematic illustration of independent circular phase rotation used to generate one panel-null permutation. Each taxon’s observed seven-component phase profile is retained intact but circularly shifted relative to the fixed DeepCoil2-anchored a–g coordinate system. Consequently, within-taxon profile shape, amplitude and cyclic ordering are preserved, whereas common phase registration among taxa is disrupted. Repeating this transformation independently across taxa generates panels representing the null hypothesis of structured individual phase profiles without privileged cross-taxon alignment. The observed cross-taxon similarity is compared with the distribution of similarity scores obtained from 9,999 such randomized panels.

The panel analysis therefore asks a different question from the individual-taxon tests:

Do the taxa exhibit greater similarity in their heptad-phase contact organization than would be expected if their phase relationships were independently arbitrary?

Observed panel statistic

For the illustrated 26-taxon analysis:

$$\text{Observed mean cross-taxon phase similarity} = 0.6451$$

This statistic summarizes the degree to which the taxon-level phase patterns exhibit a common alignment.

A value near zero would indicate little systematic cross-taxon agreement under this similarity measure, whereas a substantially positive value indicates that the taxa tend to share aspects of their phase organization.

Importantly, this does not require every taxon to possess an identical contact profile.

Taxa may differ substantially in individual residue contacts, contact amplitudes, temporal behaviour, and the relative occupancy of particular phases while nevertheless retaining a detectable common organization when expressed in the same seven-phase coordinate system.

Leave-one-out consensus

The cross-taxon similarity statistic is calculated using a leave-one-out (LOO) consensus. Each taxon is evaluated against a consensus phase pattern constructed from the remaining taxa, with the taxon being scored excluded from that consensus.

This prevents a taxon from contributing to the reference pattern against which its own similarity is measured. In other words, a taxon cannot increase its apparent agreement with the panel simply by helping to define the panel consensus.

The procedure is repeated in turn for every taxon, providing a cross-taxon similarity measure in which each comparison is made against information derived independently from the taxon being evaluated.

What the panel null preserves and what it randomizes

The panel null is designed to test shared phase alignment, rather than whether individual taxa possess structured contact profiles.

For each taxon, HARP begins with its observed seven-component phase profile describing the trajectory-derived contact distribution across the DeepCoil2-anchored heptad coordinate system.

During one panel-null permutation, each taxon is assigned an independent circular phase rotation. The values within that taxon's phase profile are therefore retained, together with their cyclic ordering, but their registration relative to the fixed a–g coordinate system is shifted.

Thus the null transformation does not replace the observed taxon profiles with random noise and does not erase their internal structure. A taxon with a strongly non-uniform phase profile remains strongly non-uniform after rotation.

What is randomized is the relative phase alignment between taxa.

Consequently, every null panel asks:

If all taxa retained their observed internal phase architecture, but their orientations within heptad space were independently arbitrary, how much cross-taxon phase similarity would arise by chance?

The observed panel statistic is then compared with the distribution obtained from repeated independently rotated panels.

This distinction is fundamental: within-taxon structure survives the null transformation; cross-taxon registration does not.

Construction of the null distribution

For each permutation, every taxon is independently circularly rotated and the cross-taxon similarity statistic recalculated. Repeating this procedure 9,999 times generates the reference distribution shown in `panel_null.png`, against which the observed, unrotated panel statistic is compared.

Interpreting the histogram

The blue histogram represents the cross-taxon similarity scores obtained from the 9,999 randomized panels.

The null distribution is centred very close to zero:

$$\text{Null mean} = -0.0005$$

with:

$$\text{Null SD} = 0.1510$$

and a:

$$\text{Null 95th percentile} = 0.2250$$

The dotted vertical line marks this 95th percentile. Thus, 95% of the randomized panel scores fall at or below approximately 0.225.

The dashed vertical line represents the actual, unrandomized 26-taxon result:

$$\text{Observed} = 0.6451$$

The observed value lies far beyond the main body of the null distribution and substantially beyond its 95th percentile.

Permutation probability

None of the 9,999 randomized panels produced a statistic as extreme as the observed result.

Using the plus-one empirical permutation convention, the resulting probability is:

$$\text{Permutation } p = 0.0001$$

This should be interpreted as evidence that a cross-taxon phase similarity at least this extreme is highly unusual under the specific panel null model implemented by HARP.

It should not be interpreted as a literal probability that the biological hypothesis is 99.99% correct.

Relationship to Figures 01–06

The individual taxon figures and the panel analysis address different levels of the problem.

Analysis	Principal question
01 Contact profile	Where does MG contact MyhT in this taxon?
02 Phase occupancy	How are those contacts distributed across DC2 phases?
03 Temporal persistence	Does that phase organization persist through the trajectory?
04–06 Taxon null tests	Is the observed organization unusual under the corresponding taxon-level null models?
Panel null	Is there a shared phase organization across the taxon panel that exceeds random cross-taxon alignment?

The panel analysis is therefore not an amalgamation of Figures 01–06. It is a higher-order test built from taxon-level phase information, using a leave-one-out consensus so that each taxon is evaluated against a shared phase pattern constructed without its own contribution.

What the panel result does not establish

The panel result does not demonstrate that all 26 taxa use an identical molecular interface.

It does not mean that every taxon contributes equally strongly to the observed coherence.

It does not establish that every individual taxon is independently statistically significant.

It also does not demonstrate that the DC2 register or structural models are biologically correct merely because the panel statistic is significant.

Rather, the result addresses a deliberately narrower question:

Given the trajectory-derived contact patterns and the independently assigned DC2 heptad coordinate systems, is the degree of shared phase organization across the analysed taxa greater than expected when their relative phase alignments are independently randomized?

For the illustrated 26-taxon HARP analysis, the observed result provides strong evidence for cross-taxon phase organisation under this null model.

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